Decoder decay and its correction over years in intracortical brain–computer interfaces

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Abstract

Implanted brain–computer interfaces can restore communication and movement after paralysis, but their decoders lose accuracy as recordings change over days and years. How fast this happens, and which corrections work, have not been measured consistently across individuals. Here, we analyzed public long-term recordings from three monkeys and 14 people with one pipeline. A fixed decoder lost half of its accuracy within one day to four months, depending on the individual, and weak regularization exaggerated this decay. In people, about 100 labeled trials restored most of the accuracy with any well-regularized refit, whereas label-free alignment added little beyond rescaling each channel. Corrections recovered far more when each channel kept its identity, and per-channel corrections tracked changes in each electrode's firing. Brief electrode silences matched session-to-session fluctuation, but silences longer than a month were real and often reversed. These results provide evidence-based baselines for building durable interfaces.

Keywords: brain–computer interface, intracortical recording, Utah array, neural drift, decoder recalibration, electrode stability

Introduction

Intracortical brain–computer interfaces (BCIs) record from small populations of neurons and turn their activity into commands. They have allowed people with paralysis

to move robotic arms and computer cursors[1–4], to type by attempted handwriting[5] and to speak through a decoded voice[6–8]. Recent work describes daily home use over years[9]. For these devices to become everyday tools, their decoders must stay accurate with little effort from the user.

The recorded signals are not stable. Within a day, signals shift as electrodes move or neurons change their firing[10]. Across days, the activity seen by each electrode changes[11–13], and over months and years, arrays gradually lose the ability to record spikes[14–17]. Decoders therefore lose accuracy over time[11,18–20], and most systems recalibrate often, which takes time away from the user[21,22]. Many methods aim to reduce this cost. Decoders can be trained on data from many days[23], updated during use[20,21,24,25], or given input layers learned for each day[5,26]. Neural activity can be aligned to a reference day without new labels[27–31], and monitoring scores have been proposed to signal when recalibration is needed[32].

Building on these approaches requires answers to three practical questions. How fast do decoders decay in different individuals, and how much of the measured decay comes from the decoder itself? Which corrections recover the loss, and with how much labeled data? And what happens at the electrodes over the years? Answers have come mostly from single datasets, spans of weeks and different baselines[33], which makes results hard to compare. In one participant, for example, a decoder whose channel means were updated still failed after months, whereas adjusting its weights recovered it[20]. Whether such results hold across individuals and species is not known.

Public long-term datasets now allow these questions to be asked across individuals with one method. They include three years of finger decoding in one monkey[19] and reaching recordings from two monkeys in another laboratory[28,34]. They also include closed-loop cursor sessions from nine BrainGate participants, spanning up to seven years, recordings from 20 arrays in 14 participants[17], and closed-loop sessions with fixed decoders[32].

Here, we analyzed these datasets with a single pipeline. We trained decoders on one day, tested them on later days and applied a ladder of corrections. Figure 1 shows the problem in one participant. Over seven years, one array lost about half of its electrodes with spikes, and the remaining spikes shrank by about half. A decoder trained 478 days earlier failed until each channel was re-weighted. Decay rates differed about a hundredfold between individuals, and weak regularization exaggerated decay. In people, about 100 labeled trials restored most of the accuracy with any well-regularized refit, and label-free alignment added little beyond rescaling each channel. Corrections recovered far more when channels kept their identity, and the per-channel corrections tracked each electrode's changing activity. At the electrode level, brief silences matched session-to-session fluctuation, whereas silences lasting over a month were real and often reversed. We release the code, a calibrated simulator and all derived results.

Results

Datasets and approach

We assembled four public datasets (Fig. 2a and Supplementary Table 1). Decoder analyses used nine individuals. Monkey N was recorded over 1,242 days with Utah

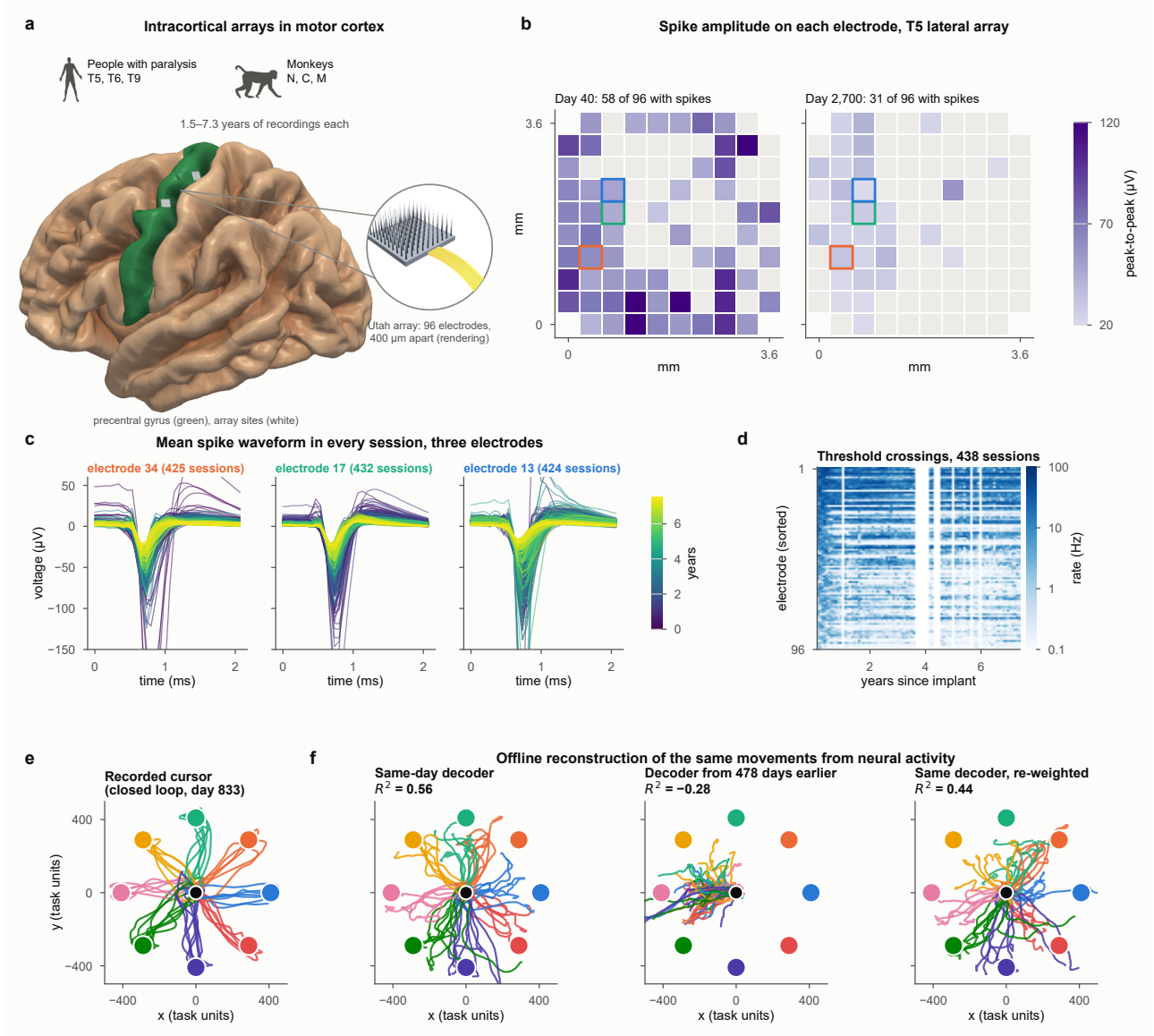


Fig. 1 Recordings change over years, and decoders decay with them. a, Approximate array sites (white) on the precentral gyrus (green) of a standard cortical surface, and a rendered Utah array. **b**, Spike amplitude on each electrode of T5's lateral array, 7.3 years apart. Grey cells had no spikes; outlines mark the electrodes in **c**. **c**, Mean spike waveforms of three electrodes in every session, coloured by time since implant. **d**, Threshold-crossing rate of every electrode on the same array, in 30-day bins. **e**, Cursor paths in a closed-loop session of T5. Colours mark targets. **f**, The same movements reconstructed offline by three decoders. The decoder from day 355 fails until each channel is re-weighted (Methods).

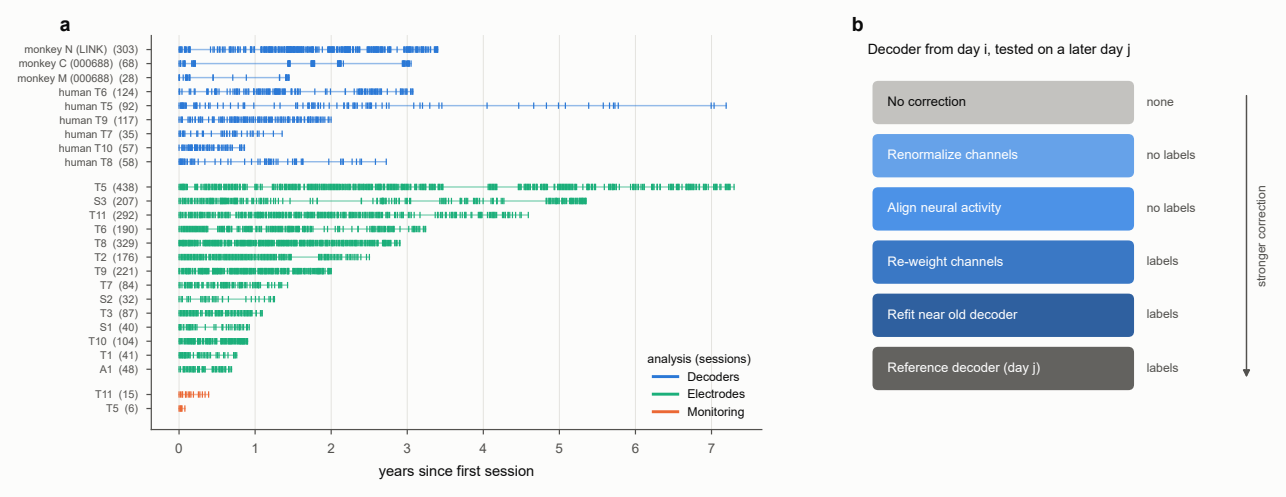


Fig. 2 Datasets and the correction ladder. a, Sessions analyzed, by individual. Each tick is one session. Numbers give sessions per analysis. **b**, The correction ladder. A decoder trained on day i is tested on a later day j after increasingly strong corrections. A decoder trained on day j is the reference.

arrays in motor cortex during a two-finger task [19,35]. Monkeys C and M were recorded over three and 1.5 years during reaching [28,34]. BrainGate participants T6, T5, T9, T7, T10 and T8 contributed 35–124 closed-loop cursor sessions each, over 0.9–7.2 years (483 sessions in total; Supplementary Table 14) [17]. Two further participants, T2 and T3, were excluded because their decoders did not work even on the day they were trained (Methods). Electrode analyses used all 20 arrays from 14 BrainGate participants (2,289 sessions) and monkey N. Monitoring analyses used closed-loop sessions in which participants T11 and T5 controlled a cursor with a fixed decoder [32].

For each individual, we trained a linear decoder on one session and tested it on later sessions at gaps of 1 to 480 days. Before testing, every channel of the later session was z-scored with that session's own mean and standard deviation. This step, renormalization, needs no labels and equalizes each channel's scale; the result is the renormalized fixed decoder. We then applied a ladder of corrections (Fig. 2b and Methods). Label-free corrections updated channel means or aligned neural activity to the training day. Corrections that use labeled trials from the later day either re-weighted each channel's contribution to the otherwise unchanged decoder, or refitted the decoder while shrinking it toward the old one. A decoder trained on the later day served as the reference. We report retention, the ratio of a decoder's accuracy (coefficient of determination, R^2) to that of the reference on the same test trials.

Decay rates differ about a hundredfold between individuals

Without any correction, decoders failed: reusing the training day's normalization gave median R^2 below zero at most gaps (Supplementary Table 2). We therefore use the renormalized fixed decoder as the baseline.

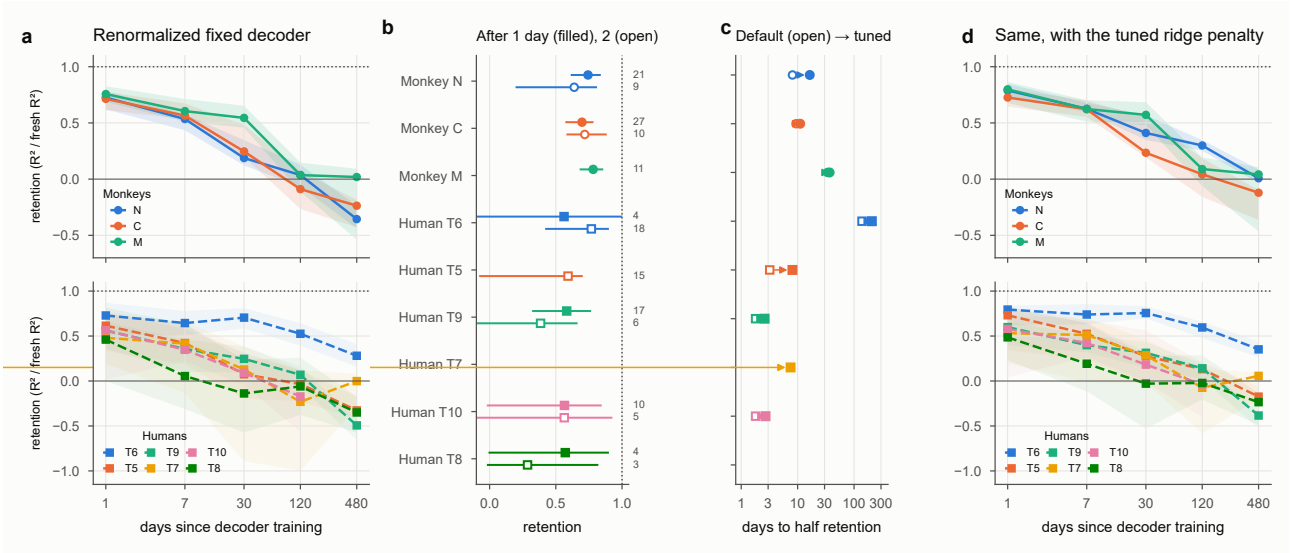


Fig. 3 Decoders decay at rates that differ between individuals. a, Retention of the renormalized fixed decoder, the R^2 of the old decoder divided by that of the reference decoder, against days since training, for monkeys (top, circles) and humans (bottom, squares). Bands show 95% confidence intervals. **b**, Retention after exactly one day (filled) and two days (open). Numbers give session pairs. **c**, Days until retention falls to one half, with the default ridge penalty (open) and the tuned penalty (filled). **d**, As in **a**, with the tuned penalty.

A single night already cost accuracy (Fig. 3a,b). For one-day gaps, retention was 0.74 in monkey N (95% confidence interval 0.62–0.83; $n = 21$ pairs), 0.70 in monkey C (0.58–0.78; $n = 27$) and 0.78 in monkey M (0.69–0.85; $n = 11$). In participants T9 and T10, which had enough one-day pairs, it was 0.58 (0.33–0.76; $n = 17$) and 0.56 (−0.01–0.84; $n = 10$). One-day pairs were rare in the other participants (Supplementary Table 2).

Longer-term decay differed far more. Retention fell to one half within a day in T7 and T8, within 1.8 days in T9 and T10 and within 3.2 days in T5 (95% confidence interval <1–8.8). It took 8.1 days in monkey N (3.7–10), 9.4 in monkey C (8.0–24), 34 in monkey M (17–45) and 138 in T6 (<1–251; Fig. 3a,c). In monkey N and all six participants, the reference decoder did not decline from short to long gaps (Supplementary Table 2). The decline therefore reflects a growing mismatch between decoder and day, not poorer recordings.

Weak regularization exaggerates decay

Part of the measured decay came from the decoder rather than the recordings. We had adopted the ridge penalty used in the original monkey N study [19] ($\alpha = 0.1$; $\alpha = 1$ for the sparser spike counts of monkeys C and M). With a stronger penalty ($\alpha = 10^4$), the time to half retention at least doubled in monkey N (8.1 to 16 days), T5 (3.2 to 8.1 days) and T7 (under one day to 7.5 days). It grew by about half in T6, T9 and T10 and changed little in monkeys C and M and in T8 (Fig. 3c,d). Per training session,

absolute R^2 on later days rose in all nine individuals, by medians of 0.003 to 0.061 (two-sided Wilcoxon signed-rank tests, $P < 0.05$ in 9 of 9). Reports of decoder decay therefore depend on how strongly the decoder was regularized, and we use the tuned penalty for the remaining decoder analyses unless stated.

Per-channel rescaling outperforms label-free alignment

Several methods realign neural activity to a reference day without labels[27–31]. Their benefits were usually measured against a fixed decoder or against one whose channel means alone were updated[20,30]. We compared such corrections with renormalization (Fig. 4a and Supplementary Table 3).

Updating channel means alone was worse than renormalization in every individual; in monkey N the difference was 0.024 in R^2 (two-sided Wilcoxon signed-rank test over training sessions, $P = 1.0 \times 10^{-10}$). Rotating the dominant neural subspace onto the training day by orthogonal Procrustes analysis[36] did not close this gap. Applied after updating means, it lowered R^2 further in monkey N (by -0.063; $P = 3.6 \times 10^{-11}$). Applied after renormalization, it lowered R^2 slightly in every individual (median changes -0.063 to -0.004; $P < 0.05$ in 8 of 9). Matching the full channel covariance (CORAL[37]) did not improve on renormalization in any monkey (monkey N: difference 0.000, $P = 0.95$; slightly worse in monkey C, difference -0.003, $P = 9.2 \times 10^{-5}$) and was worse in all six participants. A faithful implementation of the factor-analysis stabilizer of Degenhart and colleagues[27] was worse than renormalization in all three monkeys and in T6, T9, T10 and T8, did not differ in T7, and was nominally better in T5 (by 0.018; $P = 0.038$, not significant after Holm correction over the nine individuals). That method was designed for closed-loop use with a latent decoder, and offline tests may understate it. Methods that align latent dynamics with labeled data[28] or adversarial networks[29–31] were not tested. Among the corrections we tested, however, the label-free step that mattered was rescaling each channel, and gains reported for alignment over mean-only baselines[20] may largely reflect this step.

Label-free monitoring adds little beyond elapsed time

A monitor that warns when a decoder is failing could make recalibration less frequent. MINDFUL scores instability as the divergence between distributions of neural features, or of decoder outputs, on a test day and a reference day[32]. Such scores correlate with decoder error, but both grow with time since calibration.

Offline in monkey N (537 session pairs), elapsed time alone correlated with decoder accuracy (Spearman $|\rho| = 0.67$; Fig. 4b). Neural divergence correlated more weakly ($|\rho| = 0.34$), and its partial correlation fell to 0.16 after controlling for elapsed time; output divergence behaved the same way. Adding both divergences to elapsed time slightly increased the cross-validated error of predicted R^2 (0.090 versus 0.085). A pilot on the first year of monkey N tested a broader set of label-free statistics, and none improved on elapsed time (Supplementary Note 1). This was not because the remaining variation was noise: day-to-day performance beyond the time trend was reliable (split-half reliability 0.84).

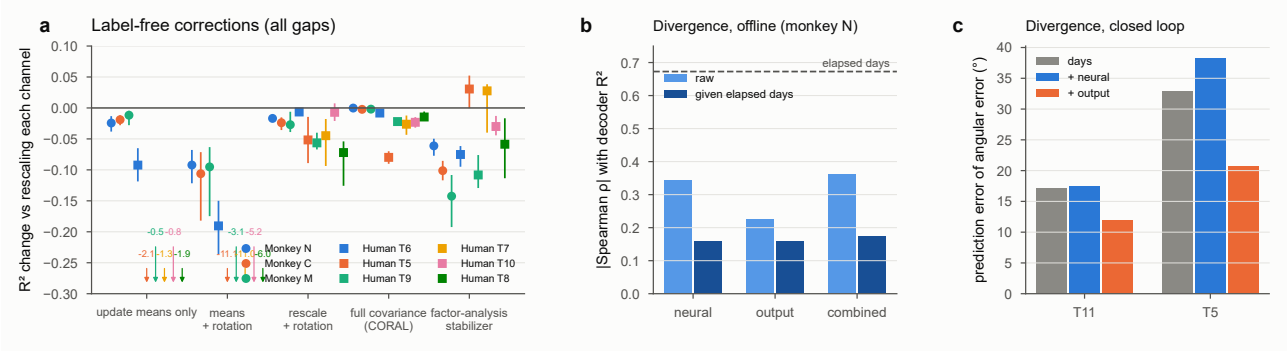


Fig. 4 Label-free corrections and monitoring add little. a, Change in R^2 for five label-free corrections, relative to renormalization. Points show medians over training sessions; bars show 95% confidence intervals. Values below the axis are printed. **b**, Offline monitoring in monkey N. Bars show how strongly each divergence score correlates with decoder accuracy, before (light) and after (dark) removing the effect of elapsed time. The dashed line marks elapsed time alone. **c**, Closed-loop monitoring in the public MINDFUL data. Bars show the error in predicting cursor angular error from elapsed days alone, or with a divergence score added.

We then re-analyzed the public MINDFUL data, in which T11 and T5 controlled a cursor in closed loop with a fixed decoder (Fig. 4c and Supplementary Table 4). In T11 (15 days), neural divergence correlated with angular error across 60-s windows ($r = 0.82$), consistent with the original report[32]. Most of this association was shared with elapsed time (partial $r = 0.37$), and adding neural divergence did not improve leave-one-day-out prediction of error (17.5° versus 17.1°). Output divergence kept a partial correlation of 0.73, and adding it reduced the error to 11.9°. These window-level statistics are nested within few days, and output divergence and angular error are both computed from the cursor's motion, so part of their link is mechanical. The six days available for T5 allow only a descriptive comparison. Within these limits, elapsed time was as good a guide as neural statistics.

Channel tuning is partly preserved over months

Label-free corrections left much of the loss in place. We next asked what kind of change the remaining loss reflects. For this we used re-weighting: one weight per channel in front of the otherwise unchanged decoder, fitted with labeled trials from the later day. Re-weighting lets each channel's contribution grow, shrink or vanish, but keeps the decoder's pattern across time lags and outputs. In the six participants it restored retention to 0.39–0.94 at gaps up to 480 days and recovered 12–84% of the loss (Fig. 5a and Supplementary Table 5). Recovery was lowest in T8 (12–65%) and highest in T5 and T9. In the monkeys it recovered −5–21%, 22–38% and 19–35%.

This difference between species mostly reflects what was decoded, not how the recordings changed. A two-dimensional target, such as the cursor-to-target direction decoded for the participants, can be fitted from many channel weights even when the decoder's per-channel patterns are wrong. To test this, we re-weighted decoders whose channels had been randomly scrambled (Fig. 5b and Supplementary Table 5).

In the participants, scrambled decoders reached a retention of 0.33–0.77, against 0.52–0.89 for correctly assigned ones. In T5 the two were close (0.77 and 0.83). In the monkeys, whose decoders predicted four kinematic signals, scrambled decoders could not be rescued (retention −0.19, 0.02 and 0.01). Decoding hand-movement direction in monkeys C and M raised the share of the loss recovered to 36–60% and 30–48% (Supplementary Table 7), but did not reproduce the human setting.

Channel identity nevertheless mattered in every individual. Correctly assigned decoders always beat scrambled ones, and the difference was largest when weights could not change sign, so that scrambled channels could not be reversed. With non-negative weights, retention in the participants was 0.38–0.89 with correct channels and 0.10–0.57 with scrambled ones. In the monkeys it was 0.49 versus −0.17, 0.59 versus 0.01 and 0.67 versus 0.00. Constraining the weights to be non-negative cost at most about 0.1 in retention with correct channels. Drift therefore preserves much of each channel's tuning pattern.

The per-channel weights also related to measured changes at each electrode (Fig. 5c and Supplementary Table 5). For each pair, we compared the fitted weights with the change in each electrode's threshold-crossing rate between the two days, recorded independently in the BrainGate yield data. Channels whose spiking rose received larger weights in 6 of 6 participants (two-sided Wilcoxon signed-rank tests over training sessions, $P < 0.05$; 6 after Holm correction across participants). The median Spearman correlation per training session was up to 0.11 and was positive in up to 92% of sessions, against near zero after shuffling channels. Channels whose impedance rose received smaller weights in T9, T7, T10 and T8 (median correlations −0.07 to −0.06; T9, T10 and T8 after Holm correction). These relations are weak for single channels but consistent, linking the decoder's corrections to physical changes at the array.

The results held for nonlinear decoders. We trained a recurrent network decoder, regularized so that it could not memorize the closed-loop sessions (Methods). It was as accurate as the linear decoder or slightly less accurate in people (median R^2 0.26–0.49 versus 0.26–0.53) and more accurate in monkeys. Re-weighting its input channels recovered 27–101% of its loss in the participants and 36–63%, 44–68% and 48–69% in the monkeys (Fig. 5d and Supplementary Table 6). In the monkeys this was more than for the linear decoder at every gap; in the participants the two ranges overlapped.

Finally, we measured the angle between decoders fitted on different days. Allowing one scale per channel reduced this angle by 8–37° in every individual (Supplementary Table 8). Angles that allow only a global scale therefore overstate how much decoders rotate across days[20].

A few minutes of labeled data restore decoders in people

How much labeled data does recalibration need? In the six participants, retention pooled over all gaps rose from 0.23 without labels to 0.67 with 50 labeled trials and 0.72 with 100 for a freshly trained, strongly regularized decoder (Fig. 6c and Supplementary Table 10). Refitting while shrinking toward the old decoder reached 0.56 and 0.66, and re-weighting reached 0.48 and 0.57. Movement periods lasted 1.6–3.9 s in these sessions, so 100 trials take a few minutes. Recovery was slowest in T8, where 100 trials restored a retrained decoder to 0.46, against 0.74–0.81 in the other participants

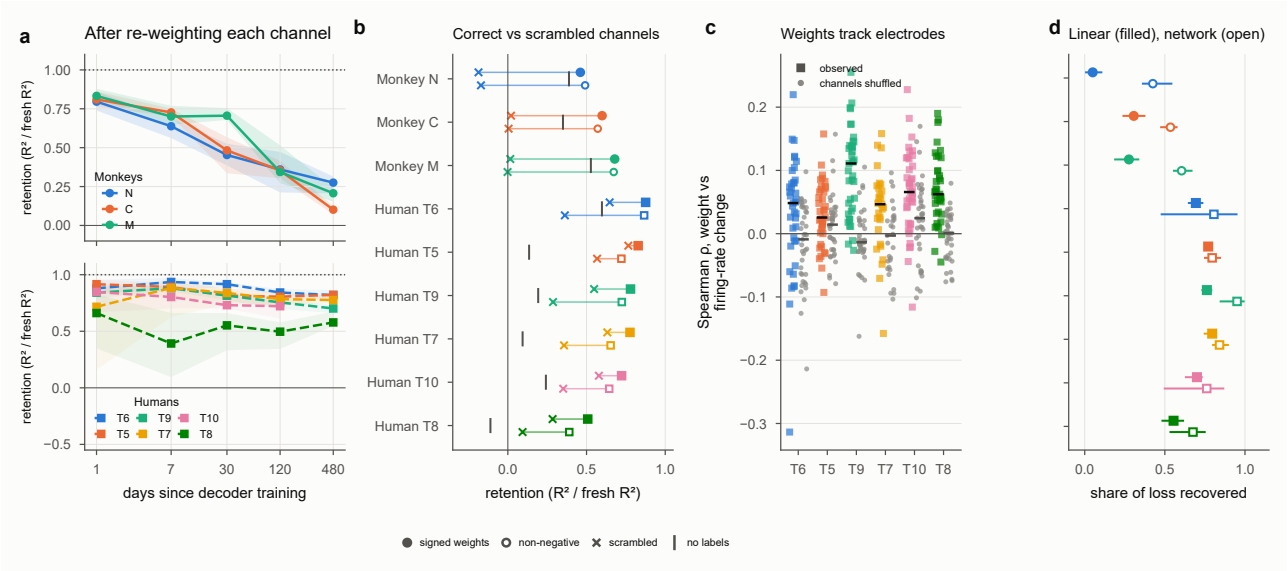


Fig. 5 Channel tuning is partly preserved over months. a, Retention after re-weighting each channel in front of the unchanged decoder, for monkeys (top) and humans (bottom). **b**, Retention after re-weighting the correct decoder (circles and squares) or a decoder with scrambled channels (crosses). Upper rows use signed weights; lower rows use non-negative weights. Ticks mark the renormalized fixed decoder. **c**, Correlation between the fitted channel weights and each electrode's change in firing rate. Each point is one training session. Grey points show the same after shuffling channels. **d**, Share of the lost accuracy recovered by re-weighting, for linear (filled) and recurrent (open) decoders. Bars show 95% confidence intervals.

(Supplementary Table 10). Re-weighting therefore indicates what changes, but it is not the most data-efficient recalibration in these sessions.

Two further choices made recalibration more reliable. First, the ridge penalty can be tuned on training-day data alone. We chose the largest penalty whose same-day accuracy stayed within 1% of the best. This rule raised cross-day R^2 in every individual without lowering same-day accuracy (Fig. 6a,b and Supplementary Table 9). Session-level median gains were 0.060 in monkey N ($P = 1.8 \times 10^{-12}$) and 0.021–0.087 in the participants ($P < 0.05$ in all nine individuals). Ordinary same-day tuning usually chose a similar penalty (Methods). Second, with very few trials, refitting can make a decoder worse. In monkey N (86 session pairs), choosing the shrinkage strength by cross-validation on 10 trials made 10.5% of recalibrations worse than no recalibration (Fig. 6d). A fixed strong shrinkage, the value that worked best on other training sessions, lowered this to 1.2%. Cross-validation was worse in 6 of 30 training sessions and the fixed value in 0 (Wilcoxon signed-rank test over training sessions, $P = 0.027$; Holm-adjusted over five data budgets, $P = 0.14$); with 20 trials the rates were 8.1% and 0.0% ($P = 0.042$; adjusted $P = 0.17$). The direction was consistent, but the difference does not survive correction for the five budgets.

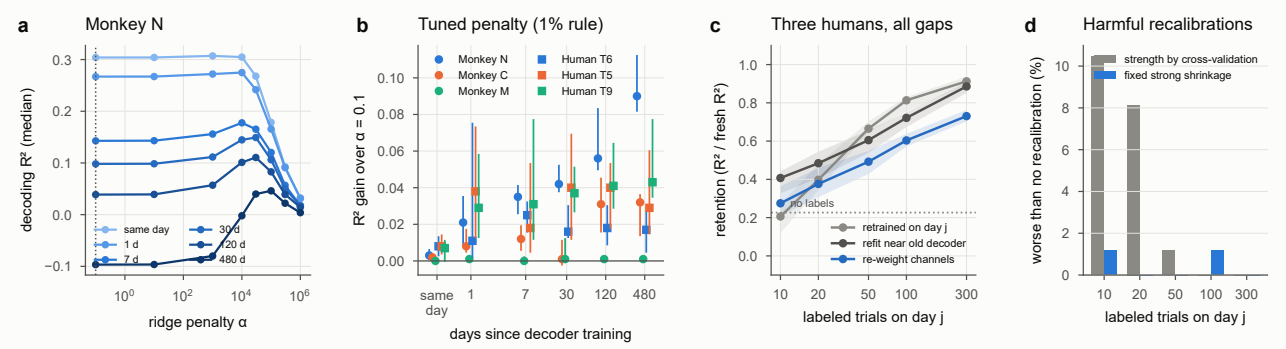


Fig. 6 Simple recipes for recalibration. a, Decoding R^2 in monkey N against the ridge penalty, on the training day (lightest) and on later days (darker). The dotted line marks $\alpha = 0.1$. **b**, Gain from the 1% rule over $\alpha = 0.1$, per individual and gap. **c**, Retention in the human participants against the number of labeled trials, for a retrained decoder, a refit shrunk toward the old decoder and re-weighting. The dotted line marks the renormalized fixed decoder. **d**, Recalibrations in monkey N that made the decoder worse than no recalibration, with the shrinkage chosen by cross-validation (grey) or fixed (blue).

Most electrode silences are transient fluctuations

We next asked how single electrodes change over years. The BrainGate yield data report threshold-crossing rates and impedance for every electrode in every session[17], and monkey N provides matching data. An electrode was active when its threshold-crossing rate exceeded 2 Hz, the definition used by BrainGate[17]. Array yield and impedance declined over years in both species (Fig. 7a,b). Impedance fell in 17 of 18 human arrays with measurements (median Spearman ρ with time -0.92). In monkey N it fell from a median of 302 kΩ in the first year to 171 kΩ in the fourth ($\rho = -0.88$). Both trends replicate earlier reports[14–17,38]. Spikes also became smaller. On T5's lateral array, 62 electrodes had spikes both in the first 200 days and after day 2,000. Their median spike amplitude fell from 55 to 29 μV (median ratio 0.51, interquartile range 0.44–0.59), and it was smaller on 61 of them ($P = 8.4 \times 10^{-12}$, Wilcoxon signed-rank test; Fig. 1b,c).

Electrodes often fell silent and later recovered[12,14], but most such events were fluctuation (Fig. 7c and Supplementary Table 11). Of 730 initially active human electrodes that stayed below threshold for at least three consecutive sessions, 80% later recovered for at least three sessions. When each electrode's session order was shuffled 200 times, which destroys any temporal process, the recovery rate was 78% (95% range 75–81%; permutation $P = 0.075$). The match also held when silence required rates below 0.5 Hz, and with a fixed threshold in microvolts for each electrode, which prevents changes in noise level from moving the threshold. Silences lasting at least 30 days were different. They were twice as common as in the shuffled data (168 versus 79 on average), and 57% of them ended in recovery, against 44% in the shuffled data (95% range 33–54%; $P = 0.010$). Recovery here means that the electrode again recorded spiking, not necessarily from the same neuron.

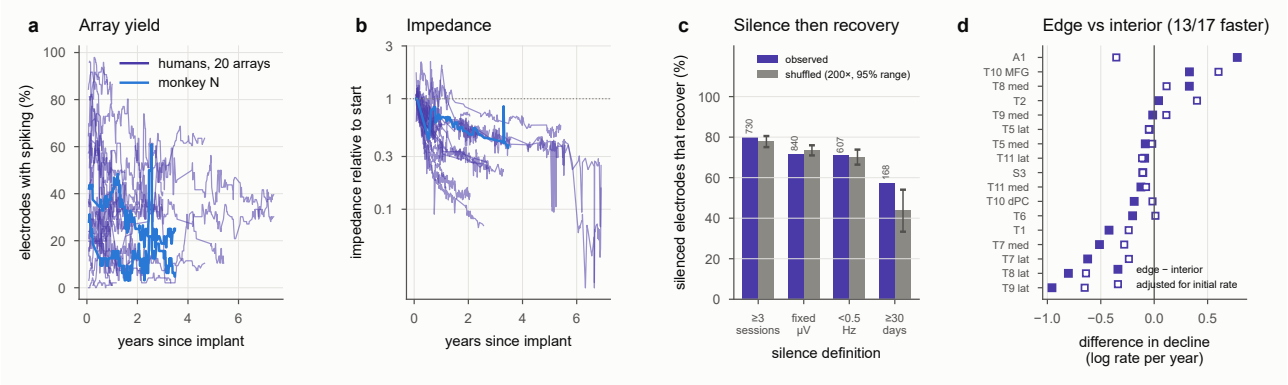


Fig. 7 Most electrode silences are transient fluctuations. a, Percentage of electrodes with spiking activity over years, for 20 human arrays (purple) and two arrays in monkey N (blue). **b**, Median impedance of each array relative to its first sessions. **c**, Percentage of silenced human electrodes that later recovered, observed (purple) and after shuffling session order 200 times (grey, with 95% range). Four definitions of silence are shown. Numbers give silenced electrodes. **d**, Difference in the yearly decline of firing rate between edge and interior electrodes, per human array, before (filled) and after (open) adjusting for initial firing rate.

Spiking declined somewhat faster at the edge of the array than in its interior (Fig. 7d). Among initially active electrodes, the edge decline was steeper in 13 of 17 human arrays with at least 10 such electrodes (Wilcoxon signed-rank test with the array as the unit, one-sided $P = 0.036$). After adjusting for each electrode's initial firing rate, the effect weakened (12 of 17 arrays; one-sided $P = 0.087$). In monkey N, edge electrodes declined faster within both arrays for threshold crossings but not for spike-band power. These trends agree with models in which micromotion strains tissue most at the array edges[39], but they are modest and partly explained by initial activity.

A calibrated simulator for testing new corrections

Methods for stable decoding are hard to compare without a common test bed[33]. We built a feature-level simulator that takes a real session and returns it as it might appear days to years later (Methods). It combines mixing of signals between channels, gradual turnover of each channel's tuning and switching of electrodes between active and silent states. Switching rates came from monkey N's electrode statistics. The mixing and turnover parameters were calibrated so that the simulated correction ladder matched monkey N's during its first 700 days. The fit had a root-mean-square error of 0.07 in retention (Fig. 8a).

We tested the simulator on the later days, which it had not seen, against two baselines (Fig. 8b). A simulator without drift was far worse (median absolute error 0.27–0.39). A stronger baseline simply reused the median retention measured in the first 700 days. This empirical curve was as good as or better than the simulator for the corrections used in calibration and for different amounts of labeled data (0.09 versus 0.12, and 0.07 versus 0.11). For corrections that the simulator was not fitted to, it was clearly better (0.06 versus 0.16). The simulator is therefore most useful

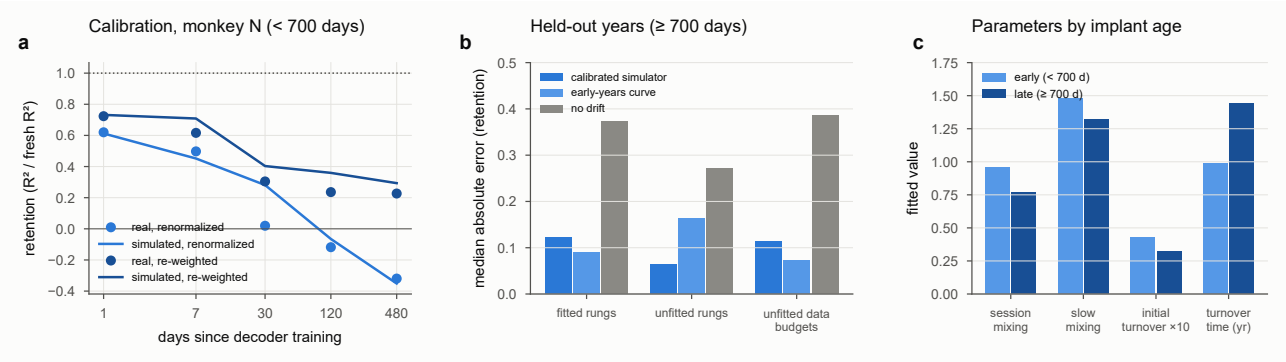


Fig. 8 A calibrated simulator for testing new corrections. **a**, Calibration on monkey N sessions before day 700. Points show real retention; lines show the simulator. **b**, Prediction error on sessions after day 700, for the simulator (dark blue), the median retention measured before day 700 (light blue) and a simulator without drift (grey). **c**, Parameters calibrated separately on early and late sessions.

for predicting how new corrections will behave. Real decoders decayed more slowly in the later years. A calibration on that period gave weaker session-to-session mixing (0.77 versus 0.96) and slower turnover (time constant 527 versus 360 days; Fig. 8c and Supplementary Table 12). Drift may therefore slow as an implant ages, at least in this monkey. Training decoders on simulated futures helped little (R^2 higher by 0.010) compared with stronger regularization (0.042) or training on several past sessions (0.056; Supplementary Table 13).

Discussion

We measured how intracortical BCIs age, at the level of decoders and of electrodes, in three monkeys and 14 people with one pipeline. Five findings stand out. Decay rates differ about a hundredfold between individuals. Measured decay depends on how strongly the decoder is regularized. In people, a few minutes of labeled data restore most of the accuracy, whereas the label-free corrections we tested add little beyond rescaling each channel. Corrections work far better when each channel keeps its identity, and they track changes at each electrode. And most brief electrode silences are noise, while long ones are real and often reverse.

These findings have direct implications for building durable interfaces. Because decay rates vary so widely, a fixed recalibration schedule is unlikely to suit all users, and schedules may need to be set per person. Because weak regularization exaggerates decay, comparisons of decoder longevity are meaningful only between well-regularized decoders. Per-channel renormalization is the appropriate baseline for evaluating label-free methods; earlier comparisons against fixed or mean-only decoders[20,30] probably overstated their benefit. For recalibration in people, any well-regularized refit with about 100 labeled trials recovered most of the accuracy in our data. When only a few trials are available, shrinking toward the previous decoder, including its intercept, made harmful updates rarer in monkey N, although that difference did not survive

correction for multiple comparisons. These are simple, inexpensive steps, and they set the bar that more sophisticated methods should clear.

The channel analyses clarify what changes. Earlier work modeled per-electrode drift as shifts in baseline over weeks[13,21], and human decoders absorb day-to-day changes with input layers learned for each day[5,26]. Other studies describe changes in the preferred directions of individual features[32] and rotation of decoder weights[20]. In our data, much of each channel's tuning pattern survived for months: decoders with correctly assigned channels consistently beat scrambled ones, and the per-channel corrections tracked each electrode's firing. At the same time, our scrambled-channel control shows that high recovery from a per-channel correction is not, by itself, evidence of per-channel drift when the decoded variable is low-dimensional. Correction-based analyses of drift, including ours, need such controls. Changes in the number or proximity of neurons near each electrode would produce per-channel changes[40,41], but our data cannot identify the cause.

The electrode analyses refine common descriptions of array decline. Activity near the yield threshold fluctuated strongly between sessions, and most short silences were no more than that. Yield counts from single sessions are therefore noisy, and channels that fall silent for a few sessions are likely to return. Silences lasting over a month were real and still reversed in more than half of cases. Hahn and colleagues analyzed yield, decoding signal quality and tuning stability in the same human data[17]. We add three analyses: silence and recovery tested against a permutation null, edge-biased spiking loss tested with the array as the unit, and a link between electrode changes and decoder corrections.

Our study has limitations. All analyses were offline; offline variability can overstate online decay[42], and in closed loop users adapt to their decoder[43]. Human decoding labels were inferred from the direction from cursor to target, and the online decoder shaped both the neural activity and the cursor path. Three of the nine BrainGate participants with decoding data were excluded. In two, the decoders did not work. In T11, recordings jumped in scale between blocks, which our session-wise normalization did not handle (Supplementary Information). Six participants and three monkeys cannot establish population rates, and the datasets differ in task, signal type, array placement and implant age. The label-free comparisons covered three classes of method and did not include latent-dynamics or adversarial alignment. The simulator is calibrated to feature-level statistics in one monkey and does not model biophysics.

Long-term public datasets made this study possible. We release our code, derived results and the calibrated simulator so that others can test methods against realistic drift. Our results suggest that keeping implanted decoders working over months depends less on elaborate corrections than on well-regularized decoders, proper per-channel normalization and short, careful recalibration.

Methods

Datasets and ethics

All data were public and de-identified. The original studies obtained ethical approval and informed consent, as described in the cited publications. We collected no new data.

Monkey N (LINK; DANDI Archive 001201[44]; ref.[19]). One rhesus macaque had Utah arrays in the hand area of primary motor cortex[35]. The release contains 312 sessions on 303 days over 1,242 days, in which the monkey moved two finger groups to targets in center-out or random-target blocks. We used the 96 released channels (64 on the medial array and 32 on the lateral array), with spike-band power and threshold crossings at -4.5 times the root-mean-square noise in 20 ms bins[45]. Outputs were the position and velocity of the two finger groups. Electrode impedance was stored on 228 days.

Monkeys C and M (DANDI Archive 000688[46]; refs.[28,34]). Both monkeys had Utah arrays in motor and premotor cortex and performed center-out and random-target reaching. Monkey C had 68 sessions from October 2013 to October 2016, and monkey M had 28 sessions from January 2014 to June 2015. We summed the spikes of all sorted units on each electrode in 20 ms bins, giving 96 (C) or 192 (M) channels. Hand position and velocity were interpolated to bin centers.

BrainGate participants (Dryad[47]; ref.[17]). The yield data contain one recording from each of 2,289 research sessions in 14 participants with 20 arrays (A1, S1–S3, T1–T3 and T5–T11). For every electrode, they list threshold-crossing rates at several thresholds (with the threshold in microvolts), impedance and grid position. Nineteen arrays were in the hand area of motor cortex and one was in the middle frontal gyrus. The decoding data contain closed-loop cursor sessions with spike-band power in 10 ms bins, cursor and target positions and the periods of attempted movement, for nine participants (T2, T3 and T5–T11). We used all sessions of each participant. Sessions, spans, arrays, trials per session and movement durations are listed in Supplementary Tables 1 and 14. All sessions of a participant were treated as one task type.

Inclusion of participants and session pairs. Retention is undefined when the reference decoder does not work. A participant therefore entered the decay and correction analyses only if the median R^2 of its reference decoder across session pairs exceeded 0.1. We fixed this rule when the first added participant (T3) failed, before the results of the other five added participants were seen. Six participants met it (T6, T5, T9, T7, T10 and T8). T2 and T3 did not (median reference R^2 -0.02 and -0.05). Their movement periods lasted a median of 15.6 s, against 1.6–3.9 s in the other participants, and neither block-wise normalization nor threshold crossings recovered a usable decoder (Supplementary Table 14). Within included individuals, session pairs whose reference R^2 was at most 0.1 were left out of retention summaries, in monkeys and humans alike. This rule was adopted after a single failed test session (reference R^2 -42.8) made bootstrap intervals meaningless; it removed 0–18% of pairs per individual and changed median retention by less than 0.05 in nearly all individuals and gaps. Results with all pairs are given in the Supplementary Information.

MINDFUL (Dryad[48]; ref.[32]). Closed-loop cursor blocks in which T11 (15 days) and T5 (6 days) used a fixed decoder, with neural features in 20 ms bins (spike-band power for T11 and threshold crossings for T5), the decoder's velocity output and the angular error of the cursor.

Features, decoders and accuracy

For monkey N and the human participants we used spike-band power (human 10 ms bins averaged in pairs). For monkeys C and M we used per-electrode spike counts. Each channel was z-scored with the mean and standard deviation of the decoding session's training segment, computed without labels (renormalization). The linear decoder was ridge regression[49] on 8 causal lags (160 ms) of the z-scored features, with an unpenalized intercept. The default penalty was $\alpha = 0.1$ for monkey N and the humans, adopted from ref.[19], and $\alpha = 1$ for monkeys C and M; the tuned penalty was $\alpha = 10^4$ for the decoder trained on day i and for the reference decoder. Outputs were finger position and velocity for monkey N, hand position and velocity for monkeys C and M, and for the humans the intended movement direction, defined as the unit vector from cursor to target during attempted movement. Accuracy was the coefficient of determination R^2, averaged over outputs and computed on held-out trials of the test session. In monkey N the first 300 of 375 trials trained the decoder, as in ref.[19]; in the other datasets the first 80% of trials did.

The recurrent decoder was a long short-term memory network[50] with 64 hidden units, input dropout[51] of 0.5 and 20-bin input windows (400 ms). It was trained with AdamW[52] (learning rate 10^{-3}, weight decay 0.01) for up to 4,000 iterations of 256 samples, with early stopping on the last 20% of training trials checked every 100 iterations. Without these measures, a 256-unit network memorized the human training sessions (training R^2 0.97, test R^2 -1.1 on one session). The network analyses used 25 training sessions per individual chosen at random with a fixed seed.

Session pairs

For each training session i and target gap $g \in \{1, 7, 30, 120, 480\}$ days, we selected the later session j of the same task type whose gap was closest to g, within 25% of g or one day, whichever was larger. The one-day set therefore contains one- and two-day gaps, which we report separately. In monkey N, 80 training sessions were drawn at random with a fixed seed; the full monkey N set (537 pairs) also includes further gaps and was used for monitoring. Pair counts are listed in Supplementary Table 2. The time to half retention was found by interpolating median retention at the target gaps linearly in log days, with 95% confidence intervals from a cluster bootstrap over training sessions.

Correction ladder

Let $\mathbf{x}_j(t) \in \mathbb{R}^C$ be the raw features of test day j, $\boldsymbol{\mu}_d$ and $\boldsymbol{\sigma}_d$ the channel means and standard deviations of day d, and f_i the decoder trained on day i.

- *No correction*: $f_i\big((\mathbf{x}_j - \boldsymbol{\mu}_i)/\boldsymbol{\sigma}_i\big)$.

- *Mean updating*: $f_i\big((\mathbf{x}_j - \boldsymbol{\mu}_j)/\boldsymbol{\sigma}_i\big)$.
- *Renormalization*: $f_i(\mathbf{z}_j)$ with $\mathbf{z}_j = (\mathbf{x}_j - \boldsymbol{\mu}_j)/\boldsymbol{\sigma}_j$.
- *Re-weighting*: f_i applied to $g_c z_{j,c}$ for each channel c, with one offset per output. The weights g_c (unconstrained in sign unless stated) were shrunk toward 1 and the offsets toward the old intercept; the penalty, scaled by the number of samples, was chosen from $\{10^{-6}, \ldots, 10^{2}\}$ by five-fold cross-validation over contiguous blocks of trials.
- *Ridge to prior*: a new decoder $(\mathbf{W}, \mathbf{b})$ fitted on the lagged features $\mathbf{H}$ of day j by minimizing

$$\|\mathbf{Y} - \mathbf{H}\mathbf{W} - \mathbf{1}\mathbf{b}^{\top}\|_F^2 + \lambda \left(\|\mathbf{W} - \mathbf{W}_i\|_F^2 + \|\mathbf{b} - \mathbf{b}_i\|_2^2\right), \tag{1}$$

  where $(\mathbf{W}_i, \mathbf{b}_i)$ is the old decoder and λ was chosen from $\{0.1, 1, \ldots, 10^{6}\}$ by cross-validation.
- *Retrained decoder*: ridge regression on day j's first n labeled trials, with α chosen from the same grid by cross-validation.
- *Reference*: a decoder trained on all of day j's training trials.

Supervised rungs used the first n labeled trials of day j ($n = 300$, or all training trials if fewer, unless stated). Retention was the ratio of a rung's R^2 to the reference R^2 on the same test trials, and the share of the loss recovered by re-weighting was $(R^2_{\text{reweight}} - R^2_{\text{renorm}})/(R^2_{\text{ref}} - R^2_{\text{renorm}})$. For the network decoder, re-weighting was a learned gain and offset per channel in front of the frozen network, and the refit was fine-tuning with an L_2 penalty toward the old weights; penalties were chosen on the last 20% of labeled trials. We also computed a full input remap fitted by L-BFGS[53]; because it can represent any linear decoder of the same form, we used it only as a calibration target for the simulator.

Channel controls. For 40 training sessions per individual (fixed seed), re-weighting was repeated with weights constrained to be non-negative (bounded least squares with the cross-validated penalty), and for a copy of the day-i decoder whose channel weights were randomly permuted, with signed and with non-negative weights.

Electrode link. For the same pairs in the human participants, the signed re-weighting weights were correlated (Spearman) across channels with the change in log threshold-crossing rate ($\log(\text{rate} + 0.1)$ at -4.5 times the noise) and in log impedance between the yield recordings of the two days, matched by electrode identity. A null correlation was computed after permuting channels.

Matched-output control. For monkeys C and M, sessions were rebuilt with a two-dimensional unit-vector target, the direction of hand velocity, on bins where hand speed exceeded the session's 25th percentile, and the ladder was run on the same pairs with the tuned penalty.

Decoder-weight angles. Each channel's block of ridge weights (8 lags by all outputs) was stacked into $\mathbf{B}_i$ and $\mathbf{B}_j$. We computed the angle between $\text{vec}(\mathbf{B}_i)$ and $\text{vec}(\mathbf{B}_j)$, which allows one global scale, and the angle between $\text{vec}(\mathbf{B}_j)$ and $\text{vec}(\text{diag}(\mathbf{s})\mathbf{B}_i)$, with each channel's scale s_c fitted by least squares.

Example in Fig. 1. Waveforms are the released robust mean waveforms (15 kHz, in microvolts) for electrodes with a threshold-crossing rate of at least 2 Hz at -4.5 times

the robust standard deviation. Amplitude is the peak-to-peak value of these waveforms. Rates in panel d are the released threshold-crossing rates at the same threshold, averaged within 30-day bins. The cortical surface was rendered with PyVista, with the precentral gyrus taken from the Desikan–Killiany atlas. Array sites were placed on its crown near the hand knob by MNI coordinates and are approximate. The array inset is a rendering from published dimensions, not a photograph. Silhouettes are from PhyloPic (CC0). For the reconstructions, decoders used the tuned penalty: one trained on T5's day 355, one trained on day 833, and the day-355 decoder re-weighted with 300 labeled trials from day 833. Paths are shown for held-out movements that started at the center. Each path integrates the decoded direction from the cursor's start position. The step size was set so that the intended direction would reach the target in the trial's duration, multiplied by a common display gain of 2 for all decoders. R^2 values are those of the analysis on the same held-out trials.

Label-free corrections

Procrustes rotation. Let $\mathbf{V}_i, \mathbf{V}_j \in \mathbb{R}^{C\times k}$ hold the top $k = 16$ principal axes of the corrected training features on each day. The orthogonal matrix $\mathbf{R}$ minimizing $\|\mathbf{V}_j\mathbf{R} - \mathbf{V}_i\|_F$ was found by Procrustes analysis[36], and inputs were mapped as $(\mathbf{I} - \mathbf{V}_j\mathbf{V}_j^\top)\mathbf{z} + \mathbf{V}_i\mathbf{R}^\top\mathbf{V}_j^\top\mathbf{z}$. We applied it to renormalized features and, separately, to mean-updated features with $\mathbf{V}_j$ estimated from those features.

CORAL. Renormalized day-j features were re-colored to match day i's covariance, $\mathbf{z}' = \mathbf{C}_i^{1/2}\mathbf{C}_j^{-1/2}\mathbf{z}$[37], with covariances shrunk 5% toward a scaled identity.

Factor-analysis stabilizer. Following ref.[27], a factor-analysis model with 10 latent factors was fitted to each day's renormalized training features, and a ridge decoder on 8 lags of the day-i posterior latent means was trained. On day j, a model fitted without labels was aligned to day i by orthogonal Procrustes analysis of the loadings of stable channels, found by iteratively keeping the 60% of channels with the smallest loading residuals; latents were then inferred from stable channels only and passed to the frozen latent decoder. The comparison baseline reused the day-i factor model on day j.

All label-free comparisons were summarized per training session over its pairs at all gaps.

Monitoring analyses

In monkey N, for every pair we computed Gaussian Kullback–Leibler divergences between days i and j of the projections of renormalized features on day i's top ten principal axes (neural divergence) and of the decoder's outputs (output divergence). We report raw Spearman correlations with R^2, partial correlations given log elapsed days and time-blocked cross-validated linear prediction of R^2. The pilot statistics (first 46–61 sessions, 283 pairs) are described in Supplementary Note 1; split-half reliability of the residual after the time trend used alternating pairs of trials and the Spearman–Brown formula. For the MINDFUL data we followed the original definitions[32]: non-overlapping 60 s windows, divergences against the first day, and the median angular error over non-excluded trials (windows with fewer than 500 valid bins skipped). We

report Pearson and partial correlations given elapsed days across windows, and leave-one-day-out linear prediction of angular error; the reference day was excluded from evaluation.

Regularization and recalibration

Ridge decoders were trained with $\alpha \in \{0.1, 10, 10^3, 10^4, 3 \times 10^4, 10^5, 3 \times 10^5, 10^6\}$ on 40 training sessions per individual (22 for monkey M) and evaluated on held-out same-day trials and on later sessions at the five gaps. The 1% rule chose the largest α whose median same-day R^2 was at least 99% of the best median. Compared with each training session's own best same-day penalty, the rule changed cross-day R^2 by a median of 0.004 in monkey N ($P = 6.9 \times 10^{-5}$) and 0.000–0.013 in the participants, and not at all in monkeys C and M. For the human data-efficiency analysis, the ladder was run for 30 training sessions per participant (fixed seed) with $n \in \{10, 20, 50, 100, 300\}$. For harmful recalibrations we used 86 monkey N pairs with default-penalty decoders; the shrinkage λ in equation (1) was chosen per pair by cross-validation on the n trials, or as the value with the best median R^2 over pairs from other training sessions, which was 10^4 for every pair with up to 50 trials. A recalibration was harmful when its R^2 was more than 0.01 below that of the renormalized fixed decoder.

Electrode statistics

Statistics were computed per array. An electrode was active in a session when its threshold-crossing rate at -4.5 times the root-mean-square noise was at least 2 Hz. An initially active electrode (mean rate of at least 2 Hz over the first three sessions) was silenced at its first run of at least three consecutive observed sessions below threshold, and recovered when it later exceeded 4 Hz for at least three consecutive sessions. Missing sessions were skipped, not imputed. Control analyses (Supplementary Table 11) varied these definitions. They used thresholds of -3.5 and -5.5 times the noise, and a fixed threshold per electrode in microvolts: its -4.5 value over the first three sessions, applied by choosing in each session the noise multiple whose microvolt threshold was closest. They also removed sessions with array-wide shifts (absolute median change in log rate above 1), required a mean waveform trough of at least 30 μV, required silence below 0.5 Hz, or required silences lasting at least 30 days. The null distribution permuted each electrode's session order independently, 200 times; we report the observed recovery fraction, the null mean and 95% range, and a one-sided permutation P value. For impedance, values of 4,000 kΩ or more were excluded and the trend was the Spearman correlation of the array median with day. For the edge effect, edge electrodes were those on the outer ring of the grid (10×10 in humans and 8×8 in monkey N), each initially active electrode's decline was the slope of $\log(\text{rate} + 0.1)$ against years, and arrays with at least 10 initially active electrodes were tested with the array as the unit. The direction was predicted from mechanical models[39], so we report one-sided tests; two-sided P values are twice as large. The adjusted analysis regressed slopes on edge membership and initial log rate within each array.

Simulator

The simulator transforms the z-scored features $\mathbf{Z} \in \mathbb{R}^{T \times C}$ of a real base session into those of a simulated session Δt days later (Supplementary Note 2). *Mixing*: $\mathbf{Z} \leftarrow \mathbf{Z}(\mathbf{I} + \mathbf{E})^{\top}$, with $\mathbf{E} = s(\Delta t)[(1-\beta)\mathbf{G} + \beta\mathbf{L}]$, where $\mathbf{G}$ is a Gaussian matrix with zero diagonal, $\mathbf{L}$ a Gaussian matrix restricted to grid neighbors, $\beta = 0.5$ and $s(\Delta t)^2 = s_0^2 + [s_\infty(1 - e^{-\Delta t/\tau_s})]^2$. *Turnover*: each channel replaces a fraction ρ_c of its signal by a unit with rotated tuning in the session's 20-dimensional latent space; ρ_c follows a beta distribution with mean $\rho_0 + (\rho_\infty - \rho_0)(1 - e^{-\Delta t/\tau_\rho})$ and concentration 5. *Silencing*: each channel active in the base session becomes silent with probability $h_{\mathrm{off}}/(h_{\mathrm{off}} + h_{\mathrm{on}})\,[1 - e^{-(h_{\mathrm{off}}+h_{\mathrm{on}})\Delta t}]$ and keeps 37% of its signal variance, the ratio of tuning strength of silent to active channels in monkey N. The switching rates ($h_{\mathrm{off}} = 0.0038$ and $h_{\mathrm{on}} = 0.0018$ per day) were measured from monkey N's activity in the calibration period. The six mixing and turnover parameters were calibrated on 12 base sessions before day 700 by minimizing the summed squared difference between simulated and real median retention for three rungs (renormalized, re-weighted and full input remap) at five gaps, with the same correction code and regularization as for real data, using a 24-point Sobol search[54] followed by Nelder–Mead[55]. Validation used 10 base sessions after day 700 and compared predicted and real medians for all rungs and for $n \in \{20, 50, 100\}$; rungs not fitted were the label-free alignments, the subspace rotation fitted with labels and the two refits. The empirical baseline predicted each late-period median by the corresponding median over pairs whose training session preceded day 700. Augmentation added eight simulated futures per training session (Δt log-uniform between 1 and 120 days) and was compared with stronger regularization, random channel dropout with gain noise[23] and training on the three previous sessions.

Statistics

Session pairs that share a training session are not independent. Confidence intervals for medians therefore came from a cluster bootstrap over training sessions[56] (2,000 resamples, percentile intervals), and paired tests used one value per training session (two-sided Wilcoxon signed-rank tests). Pairs also share test sessions, so these intervals and P values may be somewhat optimistic; we emphasize effect sizes. Harmful recalibrations were compared per training session (the share of its pairs made worse by each policy; two-sided Wilcoxon signed-rank test over training sessions), Holm-adjusted over the five data budgets[57]. Counts of significant individuals are also given after Holm correction across individuals. With six participants and three monkeys, comparisons between species are descriptive. Exact P values are reported. Analyses used Python 3.11 with NumPy, SciPy, pandas, scikit-learn and PyTorch.

Use of AI tools

An AI assistant (Claude, Anthropic) was used to write analysis code, run analyses and draft the text under the authors' direction. The authors reviewed the code, analyses and text and take full responsibility for the content.

Data availability

All data analyzed in this study are public. LINK (monkey N) is available from the DANDI Archive at https://doi.org/10.48324/dandi.001201/0.251023.2336. The recordings of monkeys C and M are available at https://doi.org/10.48324/dandi.000688/0.250122.1735. The BrainGate yield and decoding data are available from Dryad at https://doi.org/10.5061/dryad.x0k6djj1h. The MINDFUL data are available from Dryad at https://doi.org/10.5061/dryad.n2z34tn5s. All derived results underlying the figures and tables are provided in the code repository.

Code availability

All code, including the simulator with its calibrated parameters, is available at https://github.com/Imtiaj-Sajin/invasive-bci under the MIT license. A versioned archive will be deposited on Zenodo upon publication.

Acknowledgements. We thank the participants of the BrainGate clinical trials and the teams who recorded and released the LINK, DANDI 000688, BrainGate and MINDFUL datasets. This work received no specific funding.

Author contributions. All authors contributed to the study design, analysis, interpretation and writing, and approved the final manuscript.

Competing interests. The authors declare no competing interests.

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